

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: PHY101R01

Course: ENGINEERING PHYSICS

QP No. :U021R-1

Duration: 3 hours

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. Electron beam is accelerated by high potential difference in a discharge tube. Obtain an expression for the wavelength of electron.
2. The ultra-violet radiation falls on potassium surface with the wavelength of 300 nm. The work function of potassium is 2.2 eV. Calculate its kinetic energy of photoelectron.
3. Why laser transition is not possible in two level systems? Give reasons.
4. An optical fiber has core fiber index of 1.36 and relative difference in index $\Delta = 0.025$. Determine the numerical aperture and the acceptance angle of the fiber.
5. Estimate Numerical aperture and V-number of a fiber. Given $\lambda = 1000$ nm, $n_{\text{core}} = 1.46$, $n_{\text{cladding}} = 1.45$ and core radius $5\mu\text{m}$.
6. Draw E-P diagram of ferroelectric materials and briefly discuss its properties.

7. The magnetic susceptibility of material is 9.48×10^{-9} . Compute the relative permeability.
8. Differentiate polar versus non-polar molecules with examples.
9. Distinguish between elemental versus compound semiconductors.
10. At 20°C , conductivity of Ge is $2 \Omega^{-1}\text{m}^{-1}$. Find the conductivity of Ge at 40°C . $E_g = 0.72 \text{ eV}$.

PART - B

Answer any four questions

4 x 15 = 60 Marks

11. An electron is trapped in an infinitely deep potential well. Obtain an expression for energy of electron and wave function. Also discuss the results.
12. With neat energy band diagram explain the working of semiconductor diode laser.
13. Classify optical fibers based on the modes of propagation. Discuss which type of optical fiber is suitable for optical fiber communication.
14. A thin sheet of mica is placed between a parallel plate capacitor. Derive an expression for local field in a dielectric and obtain Clausius-Mossotti equation.
15. Illustrate hysteresis in a ferromagnetic material. Based on hysteresis classify hard magnetic materials versus soft magnetic materials.
16. Derive an expression of the concentration of electron in the conduction band of N-type semiconductor with diagram.

PART – C

Answer the following

1 x 20 = 20 Marks

17. (a). Discuss the concept of quantum mechanical tunneling with diagram. (4)
- (b). Estimate the wavelength emitted by Laser diode whose energy gap is 1.44 eV. (3)
- (c). If a NaCl crystal is subjected to an electric field of 1000 Vm^{-1} and the resulting polarisation is $4.3 \times 10^{-8} \text{ C/m}^2$. Calculate the relative permittivity of NaCl. (3)
- (d). Based on Hall Effect obtain an expression for carrier concentration, mobility of charge carriers and Hall coefficient of a given sample with experimental setup diagram. (10)